ART — Audit in Real Time Standard

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Canonical Definition System

ART — Audit in Real Time defines the structural conditions under which system states, events, actions, and outputs are verified continuously at the moment they occur, rather than only after they are reported, summarized, or reconstructed.

ART operates as a foundational parent standard for real-time audit logic within the OOF® system.

ART is valid only if:

- the observed event is linked to real system state
- the audit signal is captured at operational time
- the observed condition is traceable to origin
- the audit output remains verifiable and non-detached from the event it represents

Audit that depends only on delayed reporting does not meet the ART standard.

A. Standard Abstract

ART establishes audit as a live structural condition, not a retrospective document exercise.

It defines when audit may be relied upon for:

- real-time verification
- state confirmation
- event traceability
- output validation
- operational accountability

ART separates valid real-time audit from:

- delayed reporting
- reconstructed narratives
- unauditable event history
- outputs detached from source conditions

As a foundational parent standard, ART defines the audit mechanism itself.

Derived modules may later specify domain-level implementation of this mechanism.

B. Core Principle

Audit is strongest at the point of occurrence.

A system is not structurally auditable if verification begins only after the event has already passed without traceable capture.

C. Scope

ART applies to:

- digital systems
- operational workflows
- AI-assisted environments
- industrial processes
- traceable event systems
- output-generating infrastructures
- environments requiring continuous state verification

D. System Position

ART operates as:

- a real-time audit standard
- a traceability condition for system events
- a validation-linked audit layer
- a bridge between system occurrence and verifiable evidence
- a foundational parent standard for derived real-time audit modules

It does not replace broader governance.

It defines when audit becomes operationally real.

E. Structural Base

ART is based on:

- event-time capture
- state-linked verification
- traceable origin
- continuous auditability
- integrity of audit output at validated reliance

Audit must remain structurally connected to the event, state, or output it claims to verify.

F. Audit Integrity Conditions

A system implementing ART must ensure:

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1. Event-Time Capture

The relevant condition must be captured at the time it occurs.

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2. State Linkage

The audit record must remain linked to the real system state present at capture.

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3. Traceable Origin

The source of event, actor, process, or system condition must be identifiable.

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4. Verifiable Output

Audit outputs must remain verifiable against captured conditions.

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5. Condition Integrity

No audit may be treated as real-time audit if:

- the record is reconstructed without live linkage
- the source condition is missing
- the capture path is not traceable
- outputs are detached from originating events
- **Non-bypassable logic applies at validated state only.**

G. Methodology

Implementation of ART requires:

- 1. define what must be audited in real time
- 2. define the event and state capture boundary
- 3. define traceability conditions
- 4. define verification logic at the point of occurrence
- 5. define audit output structure
- 6. verify that the audit result remains linked to originating reality conditions

ART may be relied upon only after structural audit linkage
is established.

H. Invalid Conditions

A system is considered ART-invalid if:

- events are recorded only after occurrence without live capture
- audit outputs cannot be traced to originating system state
- source, actor, or process origin is not identifiable
- audit depends on interpretation rather than captured structure
- outputs are presented as verified without real-time linkage
- A timestamp alone does not create real-time audit validity.

I. System Impact

ART transforms audit from:

- retrospective review → into **live structural verification**
- reported history → into **traceable event evidence**
- document-based assurance → into **state-linked operational audit**

It establishes audit as a condition of reality-linked verification
rather than post-event interpretation.

J. Parent Standard Function

As a foundational parent standard, ART defines the governing audit
mechanism under which domain-specific modules may later operate.

ART is designed to support derived modules addressing specific audit
domains, including:

- event capture
- state verification
- output verification
- audit record integrity
- escalation under live audit conditions

These modules do not replace ART.

They extend its application.

K. Validation Layer Linkage

ART operates in connection with:

- **INTEGROS®** for integrity enforcement
- **TVL®** for validation logic
- **OBIDENITY®** for traceable origin and actor linkage
- **UCL™** for canonical meaning consistency

Audit becomes operationally meaningful only when these layers remain structurally aligned where applicable.

L. Structural Principle

Audit cannot be trusted because it exists as a record.

It can be trusted only when it remains structurally linked to the reality condition it claims to verify.

Canonical Closing Statement

If audit is not linked to reality at the time of occurrence, it is not audit in real time.

Module Architecture

→ About ART — Audit in Real Time Standard

→ Module 1 — ECM — Event Capture Module

→ Module 2 — SVM — State Verification Module

→ Module 3 — OVM — Output Verification Module

→ Module 4 — ARIM — Audit Record Integrity Module

→ Module 5 — AEM — Audit Escalation Module

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Canonical Source

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